Journal Club Pre-Session Handout

Date: January 27, 2026
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Articles:
Evered LA, Chan MTV, Han R, et al. Anaesthetic depth and delirium after major surgery: a randomized clinical trial. Br J Anaesth. 2021;127(5):704-712. doi:10.1016/j.bja.2021.07.021

Wildes TS, Mickle AM, Ben Abdallah A, et al. Effect of electroencephalography-guided anesthetic administration on postoperative delirium among older adults undergoing major surgery: the ENGAGES randomized clinical trial. JAMA. 2019;321(5):473-483. doi:10.1001/jama.2018.22005

Purpose: Review of evidence from the ENGAGES trial and the BALANCED study regarding the relationship between EEG and post-operative delirium.
Topic: EEG-targeted anesthesia and its relationship to incidence of delirium in older adults

Critical Appraisal Checklist:

	BALANCED trial	ENGAGES trial
Objective	To determine whether lighter depth of anesthesia, guided by bispectral index (BIS), reduces the incidence of postoperative delirium after major surgery in older adults.	To test whether EEG-guided anesthesia reduces the incidence of postoperative delirium compared with usual anesthesia care.
Major Points	- RCT of older adults undergoing major noncardiac surgery - Intervention = BIS-guided “lighter” anesthesia - Control = routine anesthetic care (aka no mandated BIS target) - Primary outcome = postoperative delirium	- Multicenter RCT of patients >= 60 years old - Intervention = EEG guided anesthesia - Control = usual care - Primary outcome = postoperative delirium (CAM-based assessments)
Are the results valid? - Was randomization adequate? - Was allocation concealed? - Who was blinded?	- Randomization was concealed - computer-generated, and patients and outcome assessors were blinded (not the anesthesia provider)	- Centralized randomization, allocation concealed - Patients and those assessing delirium were blinded, but not the anesthesia provider

- Were baseline characteristics similar? - Was ITT analysis performed? - Were crossovers or missing data important?	- Baseline characteristics were similar in terms of: age, comorbidities, cognitive status, surgical type and duration - ITT analysis was performed - Minimal crossover and missing data	- Baseline characteristics similar: age, frailty, baseline cognition, surgical and anesthetic characteristics - Modified ITT analysis (included all randomized patients who underwent surgery) - Some protocol nonadherence noted (ie EEG targets not always met), missing delirium data but was balanced - Had multiple imputation and sensitivity analyses which confirmed robustness
What were the results? - Effect sizes - Were results clinically meaningful or only statistically significant? - Were harms/ adverse effects reported? - Any time-course patterns? (early vs late outcomes?)	- Moderately reduced incidence of delirium in the lighter anesthesia group, with a small risk reduction (4-6%) - Does reach statistical significance, though is modest (benefit is real, but not great) - No harms (ie increase in awareness under anesthesia) or other adverse events reported	- No significant reduction in postoperative delirium (EEG guided 26% vs usual care 23%) - Difference not clinically meaningful (near zero effect size), no statistical significance - No increased intra-operative awareness or other major perioperative complications
Will these results help my patients? - Does the study population match our patients? - Are the outcomes patient-centered (survival, ambulation, pain, satisfaction)? - Any barriers in our practice (anticoagulation, resources, PACU staffing)? - How might local ERAS pathways differ?	- Older adults (ie 60-65 years), undergoing major noncardiac surgery - Patient-centered (ie delirium incidence)	- Primary population is adults >= 60 years old, undergoing major surgery, mostly noncardiac - Includes a broad, pragmatic population

PICO:

	BALANCED trial	ENGAGES trial
Population	Adults ≥60 years undergoing major noncardiac surgery; ASA III–IV; expected surgery ≥2 hours; predominantly China/Hong Kong, with sites in Australia and USA	Adults ≥60 years undergoing major surgery (cardiac and noncardiac) under general anesthesia at a US academic medical center
Intervention	Lighter general anesthesia targeted to BIS 50 using volatile anesthetics	EEG-guided anesthetic administration aimed at reducing EEG suppression and excessive anesthetic depth
Comparison	Deeper general anesthesia targeted to BIS 35	Usual anesthetic care (no mandated EEG targets)
Outcomes	Primary: Incidence of postoperative delirium (CAM-based	Primary: Postoperative delirium during POD days 1–5

	assessments) Secondary: Delirium episodes, cognitive screening (MMSE at discharge; AMTS at 30 days and 1 year), ICU admission, length of stay, myocardial infarction, mortality	(CAM/CAM-ICU) Secondary: EEG suppression, anesthetic exposure, hypotension, postoperative complications, awareness, nausea/vomiting, mortality
Time	Delirium assessed daily during hospitalization; cognitive outcomes at discharge, 30 days, and 1 year	Delirium assessed postoperative days 1–5; secondary outcomes during hospitalization and 30 days

BLUF Statements:

- BALANCED trial directly compared two anesthetic depth targets in adults >=60 years old undergoing major noncardiac surgery, predominantly in China/ Hong Kong though with other international sites as well, and found a modest reduction in delirium with lighter anesthesia.
- ENGAGES study compares EEG guidance vs usual care in >=60 year olds undergoing major surgery (both cardiac and noncardiac) at a US academic medical center and found no reduction in delirium, despite reduced anesthetic exposure.

Key Reminders:

- Lighter anesthesia may reduce delirium modestly but effects are small
- EEG guided anesthesia alone does not necessarily meaningfully prevent delirium in older patients
- Delirium prevention likely requires multifactorial strategies, not anesthesia depth alone